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Thesis

Life matters - Brain Tumor classification using deep learning

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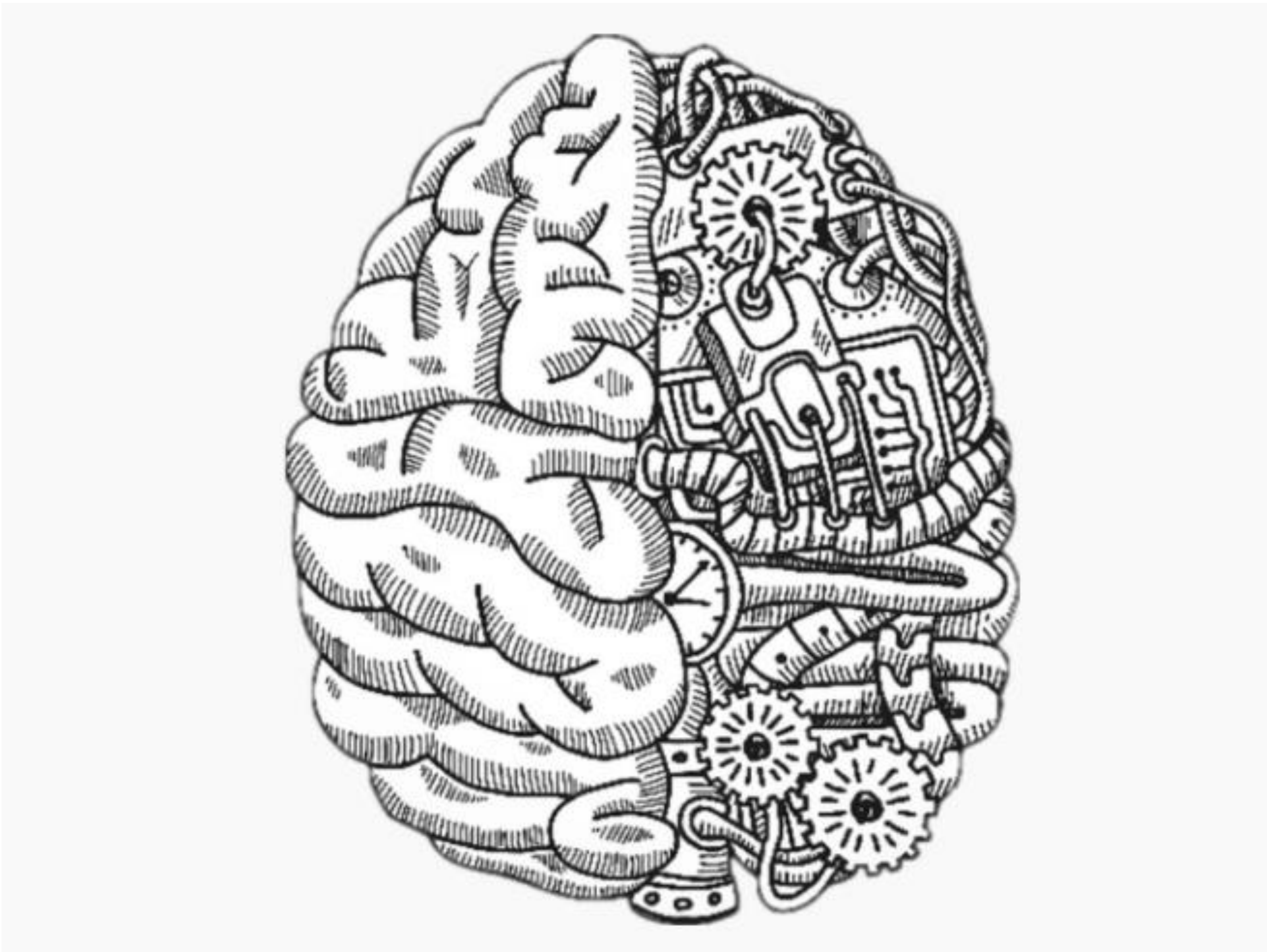

Abstract

The brain is the most sensitive organ in our body and regulates the fundamental bodily functions and characteristics. A brain tumor is a development of brain cells that multiply abnormally and

persistently [1]. Brain Tumor is the leading cause of death globally and is much common now a days. As per the National Brain Tumor Society, around 700,000 people in the United States currently have a brain tumor, and that number is expected to increase to 88,9700 by the end of 2022 [2]. Brain tumor is one of the most severe dangerous tumor diseases / disorders, it lead to a very short life expectancy when diagnosed at an upper level or very later stage and thus early discovery may help a person in mitigating the risk of life. If the tumor is detected next is identifying the type of tumor it is and what stage it is in. Additionally, grouping a brain tumor is a crucial step to perform after discovering the tumor. In this thesis, we utilized Convolutional Neural Network (CNN), one of the most popular deep learning architectures and XGBoost acting as final classifier to classify) a dataset of 3064 T1 weighted (A T1-weighted (T1W) image is a basic pulse sequence in magnetic resonance (MR) imaging and depicts differences in signal based upon intrinsic T1 relaxation time of various tissues [3].) MRI images into three categories (Meningioma, Pituitary Tumor, and Glioma). The suggested CNN + XGBoost classifier is an effective tool with an overall performance of 98.93% accuracy.

1. Introduction

Brain is the significant part of our central nervous system that controls all our functionalities through a huge number of connected neurons [4]. Brain is the most valuable organ of our body as its malfunctioning can affect entire body and it should be protected at all cost, a small damage to brain can lead to death as well in worst case.

Brain tumor is very severe disease, these tumor are produced on by the unnatural and uncontrolled growth of brain cells, and they can have fatal effects. Any malfunction in the brain cells affects the organs connected to the corresponding part of the brain, consequently damaging the functionalities of that organ. These tumor are broadly classified into primary and secondary brain or spinal cord tumor, only difference between both the categorised tumor in primary starts in the brain or spinal cord, and secondary is when the tumor started somewhere else in the body and spread to the brain [5]. The survival rate of patients post tumor detection depends on various factors namely type of tumor, its size and many other, most influential among all the factors is age. According to an article that conducted survey in Unites states found out that 5-year survival rate (people live at least 5 years) for people younger than age 15 is about 75%, For people age 15 to 39, survival rate nears 72% and survival rate for people age 40 and over is 21%.[5]. Around 400,000 people are affected by brain tumor and 120,000 people have died in the past years all over the world, as reported by World Health organization (WHO) [6]. These figures are huge and needs attention.

Initial steps are detecting brain tumor by performing brain Magnetic Resonance Imaging, it is a painless procedure that produces very clear images of the structures inside of brain. MRI uses a large magnet, radio waves and a computer to produce these detailed images. Early and proper detection can play an indispensable role in increasing the survival rate by accelerating the treatment process [7]. Typically, medical analysis, computed tomography, or magnetic imaging is used to do the analysis. Magnetic resonance imaging produces precise brain images and is one of the most often used and required methods for detecting and evaluating a brain injury. MRI is superior to other imaging modalities paired with computed tomography in medical detection systems because it provides the most accurate estimates of human soft tissues [8].

Below is the pictorial depiction of how MRI is performed for generating scans of the brain.

Figure 1: MRI San Process

Because there are so many different types and sizes of brain tumor, manual identification can be tiresome, time-consuming, and inaccurate. Expertise is required for accurate and exact identification, and challenging cases make it much harder. Therefore, in addition to human inspection, an automated procedure is required for the accurate diagnosis and classification of brain cancers. Deep learning and Convolutional neural networks can considerably speed up the entire diagnosis process, making the categorization task automated and accurate as seen in the previous research.

1.1. Background

For both men and women, brain or central nervous system cancer ranks as the eighth most common cause of mortality around the world in year 2020 [9]. Although brain tumor are not the leading cause of death globally, 40% of other cancers (such as breast or lung cancers) metastasize to the brain and become brain tumor. Despite being the gold standard for cancer detection, a biopsy has a number of drawbacks, including low sensitivity/specificity, risk when performing the biopsy, and lengthy wait times for the findings. There is a need for a non-invasive, automatic computer-aided diagnosis tool that can automatically identify and estimate the grade of a tumour accurately within a few seconds due to an increase in the sheer number of patients with brain tumor. It is classified into three types: Glioma, Meningioma, and Pituitary Tumor. To spread awareness and educate people about brain tumours, June 8 has been considered as World Brain Tumour Day since the year 2000. A brain tumour is a state where unnecessary growth of abnormal cells starts in the brain or spinal cord. In 2016, the World Health Organization graded brain tumours into four categories from low to high grade (I, II, III, and IV) based on molecular features and histology [10], [11]. Brain cancer has a minimal life expectancy when the tumour is at a higher stage [12]. Magnetic resonance imaging is the imaging modality preferred by doctors for pre- and post-treatment to estimate the structure of the tumour. According to structure of tumor these are classified in stages and categories. The type of tumor we will be focusing on are Benign Tumor, Malignant Tumor, and Pituitary Tumor.

1.2. Types of Tumor

Glioma Tumor, meningioma Tumor, and pituitary Tumor are the three other categories for brain cancers. Gliomas can develop in a number of different brain neurological systems, including the spinal column and brain pedicle. Different symptoms are brought on by different glioma kinds. Some of them include nausea, infections, headaches, and irritability. The meningioma Tumor is a tumor in the brain that stems from the meninges, and it is a somewhat frequent form of tumor that can develop in the brain [8]. A pituitary Tumor is an irregular improvement that happens in our pituitary gland. In most situations, a pituitary Tumor is an abnormal growth in the pituitary gland. The pituitary is a small gland in the brain. It is located behind the back of the nose. It makes hormones that affect many other glands and many functions in your body. Most pituitary Tumor are not cancerous (benign). They don’t spread to other parts of your body. But they can cause the pituitary to make too few or too many hormones, causing problems in the body [13]. Therefore, tumor detection strategies have worried computer researchers, specifically image processing [8].

Below is the scans for different classes of brain tumor generated from MRI in the dataset.

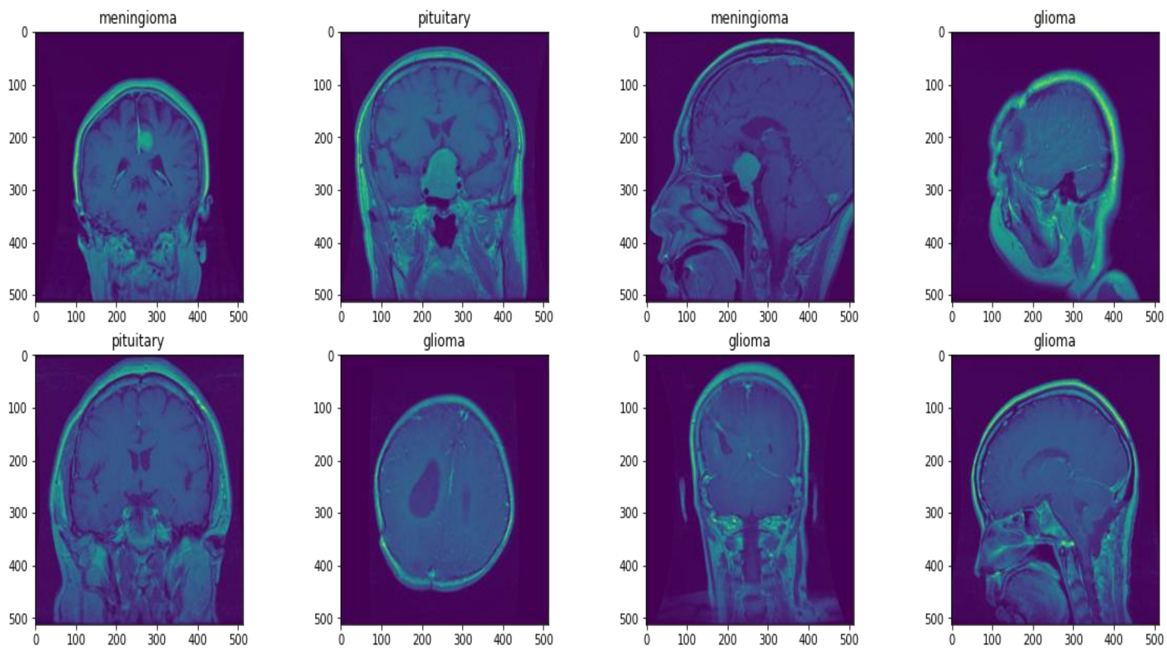

Figure 2 Brain tumor Classes

1.3. Modern Advancement

However, suppose the tumor is detected and addressed early in its development. In that case, the patient probability of recovery is quite good because treatment of a tumor is contingent upon its timely discovery [8].

Many modern technologies have taken on the critical role of automatically detecting a tumor in its early stages. Because it distinguishes the tumor from the surrounding healthy tissues, segmentation is a crucial step in the detection of tumor. The purpose of brain tumor segmentation is to locate and map the tumor regions, namely the active tumorous tissue (whether or not it is vascularized), necrotic tissue, and edema (swelling near the tumor) [12]. By comparing anomalous areas to normal tissue, abnormal spots might be found. Now, by isolating the tumor and determining its level first, the treatment process will be made simpler. It is challenging to determine a brain tumor size and resolution early in the tumor formation since exact measurements cannot be made [8].

This research proposes a novel mix of deep learning techniques, the final model consist of combining CNN and XGBoost, for classifying brain tumor. To present a general estimation of the recommended design of deep learning-based CNN and XGBoost, Cheng et al. proposed a publicly available brain tumor dataset, which is freely available online athttps://figshare.com/articles/brain_tumor_dataset/1512427/5 [14].Additionally, a comparison to other existing models was conducted to ensure that the proposed model was implemented correctly.

The thesis makes the following significant contribution:

•To provide a computerized framework for detecting brain tumor from magnetic resonance imaging pictures.

•To build a CNN approach in combination with XGBoost and a deep learning-based model.

•To demonstrate the empirical findings obtained through datasets in recognizing brain tumor with a high degree of accuracy.

1.4. Problem statement

In many medical picture applications, segmenting brain tumor is one of the most crucial and challenging processes because it typically requires a massive amount of data. Patient mobility, a short period of capture, and poorly defined soft tissue boundaries are all reasons for artefact's. There are many different tumor kinds in a wide range of sizes and shapes. They could show up in various shapes, sizes, and picture intensities. Some of them might also have an impact on the structures around, which would alter the tumor surrounding picture intensities.

In this thesis we are going to solve the problem of identification of brain tumor with an efficient and skilful method which helps in the detection of the brain tumor without any human assistance based on both traditional classifiers and CNN.

1.5. Technique selection

Deep learning models have several advantages over traditional machine learning techniques such as support vector machines, closest neighbour's, random forests, and so on. Conventional methods entail creating a feature extractor to extract predicting features from raw data/images and then organizing them into a set of optimal features that can be fed into the classifier. Furthermore, these approaches are prone to overfitting.

Therefore CNN is being utilised to overcome limitations posed by conventional machine learning models.

2. Literature Review

This section discussed about the key moments and notable research which shaped the advancement in brain tumor classification.

Machine learning and Deep Learning methods have recently been extensively used for detecting and classifying brain tumor using various imaging modalities, particularly those acquired using MRI. The most recent and related research works on the thesis topic are presented in this section. Combining images gathered from various imaging modalities while leveraging recent advances in technologies that boost the precision of detection of brain tumor in the context of artificial intelligence for image processing applications, where AI can be integrated with these imaging technologies to create an additional tool that can aid doctors in improving the precision of cancer early detection. Recently, a variety of artificial intelligence techniques have been used to categorize and identify brain tumor, including artificial neural networks, support vector machines, and convolutional neural networks (CNN) [14]–[18].

2.1. Related work

A significant amount of study has been conducted on brain tumor categorization. The majority of researchers employed traditional machine learning approaches such as nearest neighbour classifier, support vector machine classifier, random forest classifier, and others. However, these algorithms do not produce the best outcomes because traditional methods tends to work well with small dataset while the current dataset is large in size and requires significant processing as well. Recently, many imaging modalities, particularly those acquired using MRI, have been utilized to detect and grade brain cancers using machine learning (ML) and deep learning (DL) algorithms. CNN being the most common algorithm applied in recent works.

Author	Approach	Method
Rajesh et al. [21]	Machine learning	Particle Swam Optimization Neural Network (PSONN)
Bansal et al [22]	Machine learning	PSONN
Raheleh Hashemzehi et al. [8]	Machine learning	CNN + NADE
Gumaei et al. [3]	Machine learning	RELM
Kamnitsas K. et al. [23]	Machine learning	3D CNN (DeepMedic)
Nie D et al. [24]	Machine learning	SVM
Sérgio P et al. [12]	Machine learning	CNN + FE
Sonar P., [25]	Machine learning	SVM & KNN
Mehmood I et al. [26]	Machine learning	SVM
Saba T et al. [27]	Machine learning	DT, KNN, Ensemble, LDA, SVM & LR
Ginni G et al. [28]	Deep learning	KNN, RF & DT classifiers
Zacharaki et al. [29]	Machine learning	SVM
Zhan et al. [30]	Machine learning	KNN
Attique Khan et al. [31]	Deep learning	CNN
Ali Khan et al. [32]	Deep learning	CNN
Deepak et al. [33]	Machine learning	SVM and KNN

Table 1: List of Notable research on Brain Tumor Classification

Rajesh, et al. [21] developed a system that detects and classifies brain tumor. The system contains two phases that are tumor classification and feature extraction. The extraction of a feature used Rough Set Theory (RST) to extract the features and Particle Swam Optimization Neural Network (PSONN) performs the classification task. The method was trained and tested for classifying the MRI brain images as tumorous or non-tumorous. This method had 0.95%, 0.98% and 0.88% of accuracy, sensitivity and specificity, respectively.

Bansal, et al [22] used swarm ant lion classification methodology for segmenting the lesions in source brain MRI images. In this method, the authors applied canny edge detection method on source brain MRI images in order to segment the tumor regions as the segmentation process. This nonlinear approach produced significant simulation results using the swarm optimization methodology. The authors obtained 98.5% of tumor segmentation accuracy on open access BRATS 2015 dataset images.

Raheleh Hashemzehi, et al. [8] proposed a hybrid paradigm consisting of a CNN and a Neural Autoregressive Distribution Estimation to classify brain cancer from MRI images. They retrieved the characteristics and calculated the data distribution automatically. The method analysed 3064 CE-MRI images from 233 individuals, including 1426 images of Glioma, 708 images of Meningioma, and 930 images of Pituitary tumor. They demonstrated that their suggested approach has a 95 % classification accuracy.

Gumaei, et al. [4] quoted a hybrid method for feature extraction called PCA-NGIST classifying brain tumor into three categories (meningioma, glioma, pituitary) using Extreme Learning Machine classifier and achieved an accuracy of 94.23%.

Kamnitsas K, et al. [23], presented a deep multiscale 3D-CNN model named “DeepMedic” for brain cancer segmentation using MRI scans for improving brain cancer diagnosis. The model was trained on 274 MRI scans taken from BRATS 2015 dataset & evaluated on 191 scans of 94 subjects collected from BRATS 2016 dataset. By applying basic mean & standard deviation-based normalization, data augmentation & automatic kernel-based FE techniques, proposed model acquired segmentation accuracy of 89.6%.

Nie D, et al. [24], proposed a SVM based method for prognosis of brain cancer patients by inducing automatic FE of multi-modal pre-operative brain images of high-grade glioma patients. 3D-CNN, SIFT, 10-fold CV were utilized as FE techniques along with SVM model for prognosis process. Proposed model was evaluated on MRI, FMRI & DTI scans of 69 HGG patients, self-procured from Department of Radiology, University of North Carolina, Chapel Hill-USA. Proposed SVM-based prognosis model received an accuracy of 89.9%.

Sérgio P, et al. [12], proposed a DL-based segmentation method using CNN architecture for automatic segmentation of brain tumor from MRI scans of patients with gliomas. Proposed CNN model along with FE techniques i.e., bias field correction, intensity & patch-based Nyúl normalization method, data augmentation, categorical cross-entropy & Relu activation function, was evaluated on 304 MRI scans of brain tumor patients retrieved collectively from BRATS 2013 & 2015 challenge datasets and received accuracy of 0.88 (DSC). Same model was also benchmarked with previously used ML techniques & analysis results determined proposed method superior than others.

Wasule V, [25], presented an automatic ML-based methodology for incidence of malignant & benign as well as low-grade glioma (LGG) & HGG during brain tumor classification. SVM & KNN model were utilized along with FE techniques i.e., GLCM, median filter, power law transformation & morphological filtering during experimentation and evaluated on self-procured dataset of 251 MRI scans from clinical database of Sahyandri hospital, Pune. Evaluation results shows dominance of SVM-based model over KNN-based model with accuracy of 85% & 72.5% respectively.

Mehmood I, et al. [26], developed an intelligent CAD system for analysing & classifying brain MRIs to assist radiologist in correct diagnosis of brain tumor. By incorporating BoW, SURF & Volume marching cube algorithm with SVM-based classification system, proposed CAD system was trained & tested on self-procured dataset of 1100 MRI scans of 30 patients collected from Department of Radiology, Lady Reading Hospital-Peshawar and with a received accuracy of 99%, BoW driven SVM classifier-based framework outperformed regular MRI analysis tools named ITK-SNAP & 3D-Doctor.

Saba T, et al. [27], proposed a transfer-learning based model for accurate & fast classification of brain tumor by supplementing fused features to multiple ML classifiers i.e., DT, KNN, Ensemble, LDA, SVM & LR. Proposed model was supplemented with VGG, GrabCut method, LBP, HOG & entropy based fusion technique & evaluated on three public datasets named BRATS 2015, BRATS 2016 & BRATS 2017 containing 274, 274 & 285 MRI scans respectively. With a classification accuracy of 98.78%, 99.63% & 99.67% respectively, proposed model found superior than other DL-based model previously used by researchers on the above said datasets.

Ginni G, et al. [28], proposed a hybrid ensemble classifier-based model for detection & classification of brain tumor using MRI modality. This majority voting-based method utilized ensemble of KNN, RF & DT classifiers along with FE methods i.e., SWT, PCA, GLCM & Otsu's threshold during experimentation. Proposed method was evaluated on 2556 MRI images of patients retrieved from public database named TCGA-GBM and with an improved accuracy of 97.31%, proposed hybrid ensemble-based classifier outperformed other utilized traditional individual ML classifier i.e., SVM, NB, DT, NN & KNN during classification process.

For instance, Zacharaki, et al. [29] presented a technique for classification of brain tumor into various types i.e., gliomas, primary gliomas, and metastases. This study comprises of numerous stages: region-of-interest (ROI) extraction followed by feature extraction, feature selection, and classification. The classification task was performed via a support vector machine (SVM). The highest accuracy of 97.8% was seen in grade 2 gliomas versus metastasis classification, and the lowest accuracy of 75% was observed between grade 2 and grade 3 gliomas classification.

Zhan, et al. [30] also performed a similar study to grade the tumour malignancy in LGG and HGG using a multi-sequence MRI. Intensity, volume, and local binary pattern features were extracted from the multi-sequence MRI. The principal component analysis method was adopted for feature reduction. The highest classification accuracy of 87.59% was seen using the k-nearest neighbours (KNN) classifier.

Attique Khan, et al. [31] proposed a transfer learning-based DL framework for brain tumour classification. They utilised two well-known pre-trained networks, namely VGG16 and VGG19, for automated feature extraction. The popular Brain Tumour Segmentation (BraTs) dataset was used for the experiment. Further, correntropy and the partial least square-based method were used for suitable feature selection and fused in a single matrix. Finally, an extreme learning machine (ELM)-based classifier was used and the highest accuracy of 97.8%, 96.9%, 92.5% was attained on BraTs2015, BraTs2017, and BraTs2018 datasets, respectively.

Another similar study, by Hassan Ali Khan, et al. [32], proposed a DL-based brain tumour classification method between malignant and benign tumour MRI. The MRI dataset was taken from the Kaggle website. The dataset includes 155 images of malignant tumours and 98 of benign tumours. Three well-known pre-trained CNNs, VGG-16, ResNet-50, and Inception-v3 models were used in the transfer learning paradigm. The pre-trained models, VGG-16, ResNet-50, and Inception-V3 achieved 96%, 89%, and 75% classification accuracy between malignant versus benign image classification, respectively. The DL models, VGG-16, ResNet-50 and Inception-V3, show inconsistent results with respect to their layer count.

Further, Deepak, et al. [33] performed a brain tumour classification study on whole image data (MRI) for brain tumour classification using transfer learning. The proposed method was used pre-trained CNN, GoogleNet to extract features, and used support vector machine (SVM) and KNN as a classifier instead of softmax. The method classifies brain tumours into three classes, gliomas, meningioma, and pituitary tumours using figshare dataset. The proposed method recoded the highest mean accuracy of 98% using the KNN classifier.

2.2. Outline

Although there exist several techniques for brain tumor classification including conventional and non-conventional machine learning techniques, but they have certain limitations which need to be considered while working with brain tumor classification. The first limitation of these systems is their binary classification of tumor, which leaves many ambiguities for the radiologist. The reason is that classification into benign and malignant is not enough for radiologist to decide treatment and preventions for the patient. For clear and better understanding of radiologist, the classification needs to be multi-class, which classifies brain tumor into its relevant grades. Furthermore, lack of data or the unavailability of authentic/public data is also a key challenge for researchers to achieve precise results, since there are very few available datasets for brain tumor that are publicly available. To address these key limitations, we tested few other CNN models and finally proposed a deep learning-based framework which is a fusion of CNN and XGBoost to achieve effective results for multi-grade brain tumor classification.

2.3. Proposed method

Recently DL has been emerged due to its high rate of accuracy and its vast applications domain in many research areas including computer vision, image processing, authentication system, and speech recognition [34]. CNNs are feed-forward artificial neural networks encouraged by natural procedures intended to identify different patterns directly from image data. Motivated from the recent achievements of CNNs in various challenging tasks we used CNN to target the problem of multi-grade brain tumor classification, in addition XGBoost as an optimized distributed gradient boosting library designed to be highly efficient, flexible and portable. It implements machine learning algorithms under the Gradient Boosting framework. XGBoost provides a parallel tree boosting (also known as GBDT, GBM) that solve many data science problems in a fast and accurate way [35] was also utilised.

In this article, we present a novel deep learning framework to segment and classify brain tumor into three different classes using a CNN with XGBoost model. The proposed system consists of three main steps: 1) tumor segmentation, 2) data pre-processing, and 3) classification. In first step, tumor regions in the datasets are segmented. Tumor regions are segmented through a densenet121 resolves problem of information being deprecating as the number of convolution layer increases by modifying the standard CNN architecture and simplifying the connectivity pattern between layers. In a DenseNet architecture, each layer is connected directly with every other layer [36]. The layers of it explicitly support segmentation. The second step involves normalizing, flipping, rotating image. Later in the last step, data is passed to a CNN + XGBoost architecture trained on brain MR images for the final prediction of tumor grades. The detailed proposed framework is shown in Fig. 3.

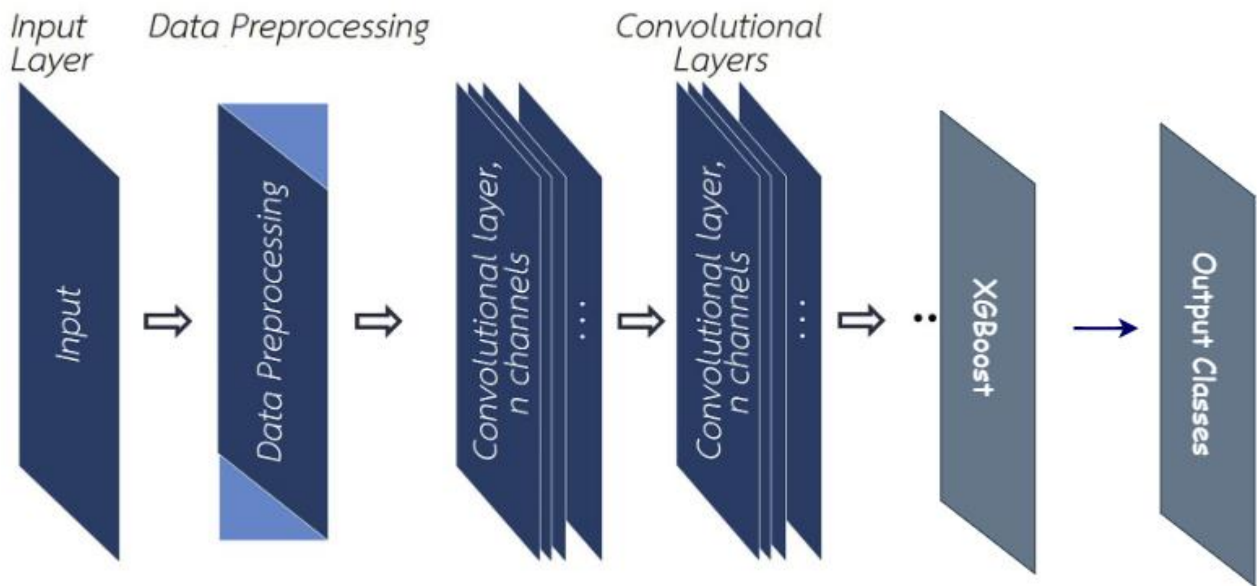

Figure 3: Framework

The architecture shows a step by step procedure followed for solving classification problem. Input layer is the first layer in the architecture which receives image as an input in our case, later the image input have been pre-processed in order to fit in the model. Pre-processing involves rescaling and normalisation. Image normalization is a process that puts multiple images into a common statistical distribution in terms of size and pixel values; however, a single image can also be normalized within itself. Typically, the procedure includes both spatial and intensity normalization. We normalized image to 224*224 pixel size. Next are the convolution layers which performs feature extraction layer by layer and generates feature map for the next layers. Finally XGBoost is used for the classification once all the features are extracted.

3. Methodology

This section explains about data, how the data was gathered, data pre-processing, and the methodology. Finding data was a difficult effort since most of the data is not made available to public. Data pre-processing was also a challenging phase because the data was not in simple text format. Considering model selection; there are numerous approaches available, but this thesis used deep learning technique.

3.1. Data Collection

Surveys, interviews, and obtaining information from approved organizations are just a few of the data collection techniques. Secondary data are those acquired through or which used by other researchers, whereas primary data are those obtained through surveys and interview scheduling. Datasets regarding this project's issue were searched on a number of authoritative medical websites and surveys. This approach of examining the dataset yielded the main conclusion that the data was a bit small meaning less number of images to be fed into the model but good enough to proceed, which is the ideal state in real life. To focus on the project's main goal of classifying brain tumor into their appropriate categories Maningioma, glioma, and pituitary brain tumor are the first two tumor for which XGBoost is employed with CNN models.

Secondary data is used in this research, as specified in the project's scope.

3.2. Dataset

In this research author has considered the brain tumor dataset suggested by Cheng, Jun, et al and is freely accessible online at https://figshare.com/articles/brain_tumor_dataset/1512427/5 [14] . The dataset has 3064 T1-weighted MRI scans from three different tumor classes: pituitary tumor, meningioma, and glioma. The number of photos for each class is shown in Table 2. Each image is in .mat format in the dataset with properties as the patient id (PID), border, tumor, tumor mask, and class label.

Summary of the dataset is listed in below table 1.

Class	**Number of Images**
Glioma Tumor	1426
Meningioma Tumor	708
Pituitary Tumor	930
Total	3064

Table 2: Summary of used dataset

3.3. Environment Setup

It takes a machine considerable amount of processing power to handle tasks like loading the dataset, pre-processing, model fitting and training especially when the data is huge or in image formats. The availability of the Graphical Processing Unit for intensive computations was the project's biggest risk. The Munster Technological University assisted with this need and gave virtual machine access to the GPU cluster. To connect to the GPU cluster, a virtual private network (VPN) gateway has been set up. Google colab, was also used, it is very useful platform if the data processing needs GPU or when the data is huge. It makes model building and fitting really easy and smooth.

Once we have the proper environment to run the model, several libraries were imported. Libraries for every aspect of data processing, like for visualisation seaborn and many more for each stage were imported and used respectively. Finally, major libraries such as Keras were used to build pre-trained models.

3.4. Methodology Used

In this thesis, we tried and tested variety of different libraries, tools and, frameworks and have compared them and chosen the one that performed best.

3.4.1. Convolutional Neural Network

The most popular deep-feed forward neural network at the moment that can handle diverse types of data inputs, such as 2D images or 1D signals, is the CNN.

The input layer, convolution layer, RELU layer, pooling layer, fully connected layer, softmax/logistic layer, and output layer are the many layers that make up CNN [37].

The data is distributed in training and testing data, with 90% of it in training and the rest 10% in testing. The structure of the suggested CNN with XGBoost architecture will be explained in the following sections.

There are libraries like keras, TF, XGBoost are utilised while working with CNN, the researcher has tried and tested as well.

Keras is an effective high-level neural network Application Programming Interface written in Python. This open-source neural network library is designed to provide fast experimentation with deep neural networks, and it can run on top of CNTK, TensorFlow, and Theano [38].

TensorFlow is a symbolic math library used for neural networks and is best suited for dataflow programming across a range of tasks. It offers multiple abstraction levels for building and training models [38].

XGBoost is a distributed gradient boosting library that has been developed to be very effective, adaptable, and portable. It uses the Gradient Boosting framework to implement machine learning algorithms. A parallel tree boosting method called XGBoost (also known as GBDT or GBM) is available to quickly and accurately address a variety of data science issues [35].

3.4.2. Proposed CNN Architecture

In this thesis, we have proposed and compared various different libraries in conjunction with CNN and CNN architecture acting individually as well. The best model observed was when CNN and XGBoost were used together. There were multiple libraries, framework, and API's were also tested during the research, apart from the above mentioned studies in order to get the most of CNN and also to increase the size of dataset basic data augmentation techniques like Rotation and Flipping are also investigated.

The architecture utilises Densely CNN called DenseNets121, there are other options available as well LeNet with 5 layers, to VGG with 19 layers and ResNets surpassing 100 even 1000 layers. In a traditional feed-forward CNN, each convolutional layer except the first one (which takes in the input), receives the output of the previous convolutional layer and produces an output feature map that is then passed on to the next convolutional layer and as the number of layers in the CNN increase, i.e. as they get deeper, which causes information to 'vanish' or get lost which reduces the ability of the network to train effectively. Thus DenseNets resolve this problem by modifying the standard CNN architecture and simplifying the connectivity pattern between layers. In a DenseNet architecture, each layer is connected directly with every other layer [36]. Densenet keeps on increasing the depth of deep convolutional networks, with which helps the classifier effectively rate the brain tumor

Below is the structure of densenet with five layers.

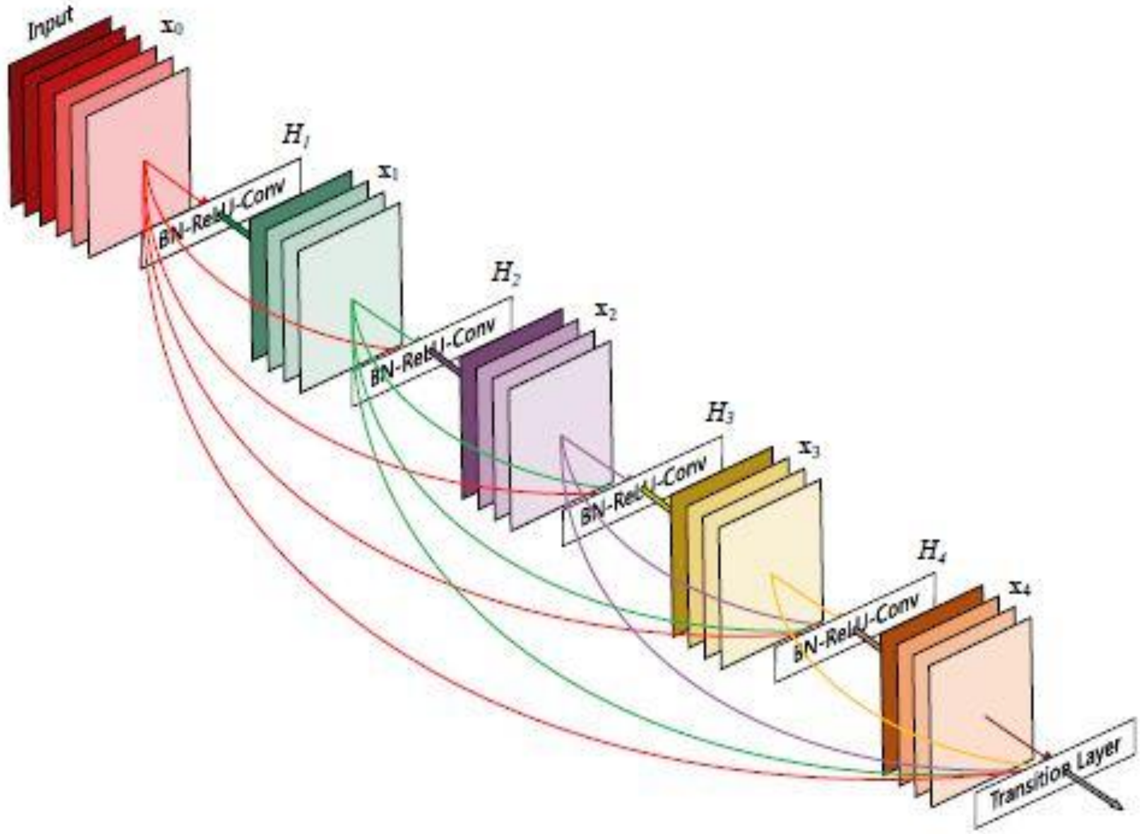

Figure 4: DenseNet with 5 layers with expansion of 4. [37]

Following the application of a composite of operations, traditional feed-forward neural networks connect the output of the layer to the following layer. An activation function, a batch normalization, and a convolution operation, or pooling layers, are all included in this composite. The equation for this would be:

$$x_l = H_l(x_{l-1})$$

Hl : Convolution layers

Xl : feature maps generated by layers

ResNets extended this behaviour including the skip connection, reformulating this equation into:

$$x_l = H_l(x_{l-1}) + x_{l-1}$$

DenseNet make the first difference with ResNets right here. DenseNet do not sum the output feature maps of the layer with the incoming feature maps but concatenate them. Consequently, the equation reshapes again into:

$$x_l = H_l([x_0, x_1, \ldots, x_{l-1}])$$

X0, x1: feature maps

When the sizes of the feature maps differ, ResNet cannot do this grouping of the feature maps. Regardless of whether the grouping is a concatenation or an addition. As a result, and similarly to ResNets, DenseNet are divided into DenseBlocks, where the size of the feature maps is constant within a block but varies between them in terms of the number of filters. Transition Layers are the layers in between them that handle the downsampling by performing batch normalization, 1x1 convolution, and 2x2 pooling layers.[36]

We are now prepared to discuss the growth rate. This channel dimension is getting bigger at every layer because we are concatenating feature maps. We can generalize for the l-th layer if we force H l to generate k feature maps each time:

$$k_l = k_0 + k * (l - 1)$$

Growth rate is indicated by the hyperparameter k. The amount of information added to the network at each tier is controlled by the growth rate.

The feature maps could be thought of as the network's data. Every layer has access to the collective knowledge and its previous feature maps. Then, in tangible k feature maps of information, each layer is adding a new piece of knowledge to this body of knowledge.[36]

Because of the highly dense number of connections on the DenseNet, the visualization gets a little bit more complex.

Below is the DenseNet-121 Architecture with all the acting layers involved in it.

DenseNet-121 Architecture

Layers	Output Size	DenseNet-121	DenseNet-169	DenseNet-201	DenseNet-264
Convolution	112×112	7×7 conv, stride 2			
Pooling	56×56	3×3 max pool, stride 2			
Dense Block (1)	56×56	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 6$	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 6$	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 6$	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 6$
Transition Layer (1)	56×56	1×1 conv			
	28×28	2×2 average pool, stride 2			
Dense Block (2)	28×28	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 12$	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 12$	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 12$	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 12$
Transition Layer (2)	28×28	1×1 conv			
	14×14	2×2 average pool, stride 2			
Dense Block (3)	14×14	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 24$	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 32$	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 48$	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 64$
Transition Layer (3)	14×14	1×1 conv			
	7×7	2×2 average pool, stride 2			
Dense Block (4)	7×7	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 16$	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 32$	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 32$	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 48$
Classification Layer	1×1	7×7 global average pool			
		1000D fully-connected, softmax			

Figure 5: Densenet-121 Architecture

When analysing the table above using the DenseNet-121 architecture, we can observe that each dense block has a different number of layers (repetition), each of which contains two convolutions: a bottleneck layer that is 1x1 in size and a convolution layer that is 3x3 in size.[36]

A 1x1 convolutional layer and a 2x2 average pooling layer with a stride of two are also present in each transition layer. Consequently, the layers at play are as follows:

1. Basic convolution layer with 64 filters of size 7X7 and a stride of 2
2. Basic pooling layer with 3x3 max pooling and a stride of 2
3. Dense Block 1 with 2 convolutions repeated 6 times
4. Transition layer 1 (1 Conv + 1 AvgPool)
5. Dense Block 2 with 2 convolutions repeated 12 times
6. Transition layer 2 (1 Conv + 1 AvgPool)
7. Dense Block 3 with 2 convolutions repeated 24 times
8. Transition layer 3 (1 Conv + 1 AvgPool)
9. Dense Block 4 with 2 convolutions repeated 16 times
10. Global Average Pooling layer- accepts all the feature maps of the network to perform classification
11. Output layer

Therefore, DenseNet-121 has the following layers:

- 1 7x7 Convolution
- 58 3x3 Convolution
- 61 1x1 Convolution

- 4 AvgPool
- 1 Fully Connected Layer

In short, DenseNet-121 has 120 Convolutions and 4 AvgPool [36].

The model architecture consists of several stacked convolutional layers to recognize input features and is able to learn the features automatically, after performing the required training of the convolutional neural network on brain tumor images, we have extracted the last layer preceding the Dense classification layers while building the proposed neural network and from it we will extract the features and train XGBoost in the last layer.

Below is the entire work flow architecture of the final model.

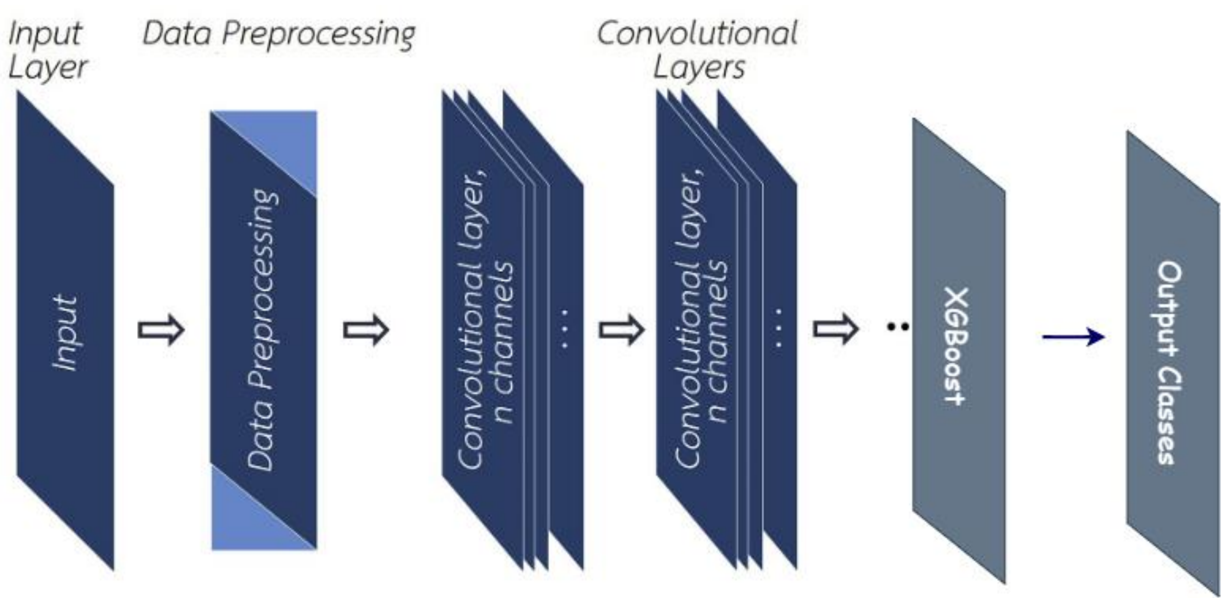

Figure 6: Model Design

In this thesis effort regularization is also investigated by using L2 regularizers by keras.

Final model have DenseNet121 layer with image input shape of (224*224*3) followed by a dropout layers with the rate of 0.8 neurons drop (The Dropout layer randomly sets input units to 0 with a frequency of rate at each step during training time, which helps prevent overfitting.). Next is GlobalAveragePooling2D layer that is used for applying global average pooling operation for spatial data. The last layer is a dense layer which implements the operation on input and return the output:

Output = activation (dot (input, kernel) + bias)

- **input** represent the input data
- **kernel** represent the weight data created by the layer
- **dot** represent numpy dot product of all input and its corresponding weights
- **bias** represent a biased value used in machine learning to optimize the model
- **Activation** (Softmax) represent the activation function.

3.5. Data pre-processing

Even after successfully obtaining all three classes of tumor images—1426 MRI scans of Gliomas, 708 MRI scans of Meningiomas, 930 MRI scans of Pituitary Tumor, - it would be crucial to choose the highest quality MRI scans from the dataset. Deshpande et al. emphasize that a select number of parameters, including the calibre of an MRI device and low image resolutions, can cause MRI picture quality to deteriorate [39]. They further assert that it would be difficult to find malignancies in low-resolution photos. High-quality resolution makes it easier to overcome obstacles in the future, which leads to the final best outcomes for administering the greatest diagnosis.

The process will comprise numerous additional steps, including image extraction and scaling (normalization) by dividing image pixels by 255. This procedure involved extracting and resizing each of the chosen photos to 224*224 pixel. This is something that would be necessary while managing photographs because it would be challenging to catch all of the images in the same shapes and sizes when they are being captured. Algorithms using images of various sizes could take longer or perhaps result in unexpected misreading's. The processing time for identical-sized images can be minimized, and distortion cases can be decreased.

All photographs would need to be resized in order to reach the appropriate size for processing. The resizing procedure must guarantee that no data on the chosen image was lost. The resized image's height and width are both kept at 224 pixels. This task of resizing has been carried out using a dense layers.

3.6. Data Visualization:

Data visualization is a crucial part of any analysis or research, it makes use of the information and present it in such a remarkable way that it's easy for readers as well to get an exact idea of what comparisons, categorisations and number facts are there in the dataset. Since in our image dataset is of secondary nature of the data and the fact that train, validation, and test sets have already been created, there is not much of visualisation required.

To display dataset information, a data visualization method is still used on the dataset. Utilizing graphs like the bar plot, the number of brain tumor photos for each of the dataset's types of brain tumor has been displayed.

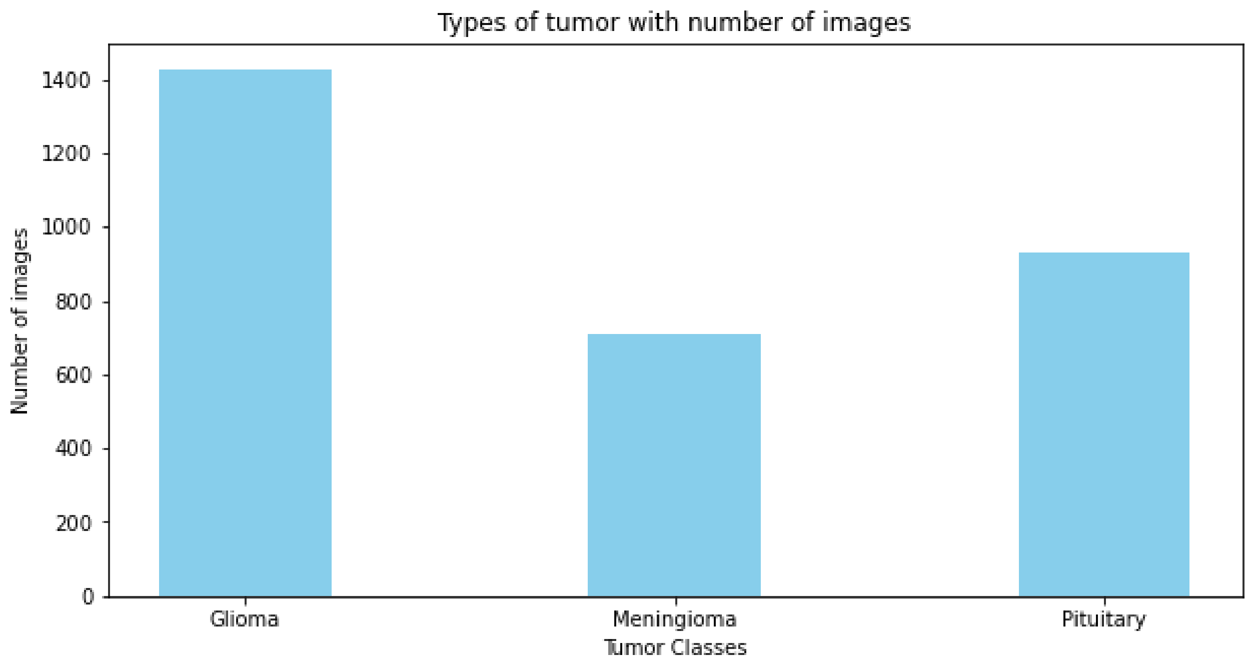

Figure 7 : Brain Tumor types and their numbers in dataset

To learn more about the tumor, photographs of other types of brain tumor have also been exhibited. The categories of each image of a brain tumor are displayed using the Python packages CV2 and Pillow. There are other available libraries too but we found these libraries easy to use.

Additionally, line graphs have been utilized to compare the CNN model's accuracy and loss on training and validation data in order to compare model performance. The Matplotlib Python package has been used to visualize these charts.

3.7. Data Augmentation:

Once the data pre-processing is completed meaning after all of the chosen images have been resized, the next stage comes is data augmentation. After being resized, each of the chosen photographs has undergone a number of processes that result in the generation of new samples while maintaining the integrity of the original data. The process for data augmentation would be followed. The final phase involved grouping all of the images into a batch of two by applying a concatenation of randomly applied flips, rotations, translations, and other related operations. Sending the data to the final CNN model for training comes next after converting the image into a batch.

Only Basic Data Augmentation (Flipping, rotation, shifting, brightness), were followed.

Below augmentations were performed on the dataset:

1. Rotation augmentations are done by rotating the image right or left on an axis between 1° and 359°. - rotation_range=0.2,
2. zoom_range=0.01,
3. horizontal_flip=True,

4. vertical_flip=True

This technique includes image transformation properties such as image flipping (horizontal and vertical), zooming range applications, and rotation by 0.2. Both training and validation imaging data of brain tumor have undergone these changes. More picture data has been produced to train a CNN model as a result of these image alterations.

Below is the before and after augmentation images of scans shown:

Before

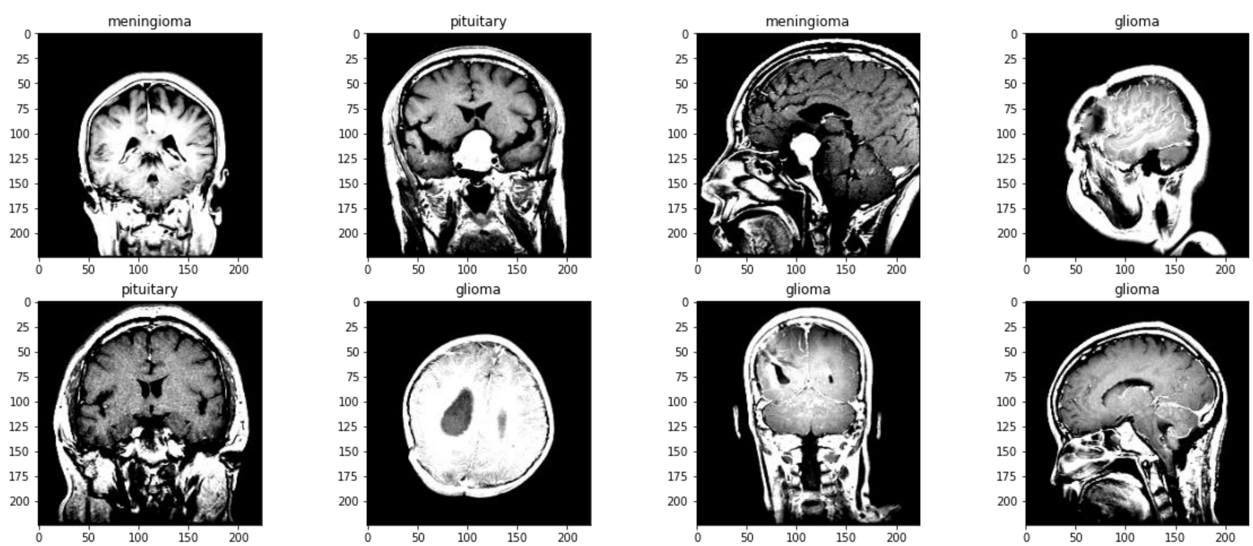

Figure 8: Original Scans

After

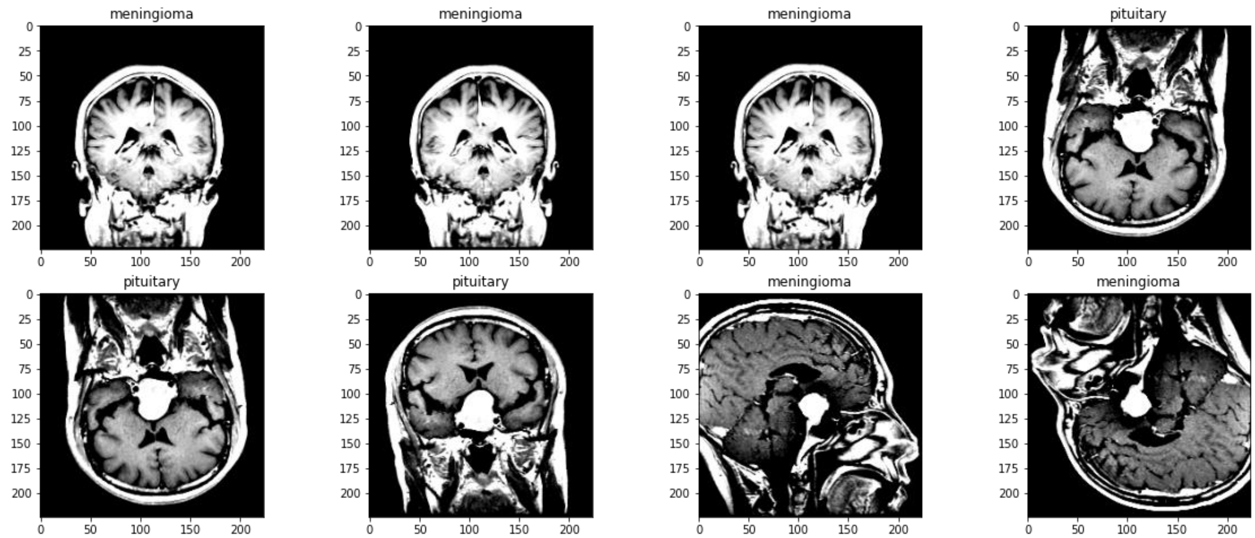

Figure 9: Augmented scans

The above augmented images in figure 9 shows images being flipped and rotated in order to feed to model. For example Meningioma images one rotated left and other flipped vertically in the figure. This enables the model to train and extract more features of a particular class of tumor with a new set of images, this results in model performing well for training as well as testing data.

3.7.1. Model Training:

The last step is CNN's involvement. Since we used the densenet121 layer, this neural network supports a number of layers, including fully connected, pooling, and convolution layers. In this step, the image and its operations are categorized into various groups**.** Regularization techniques like batch normalization, kernel initializer, kernel regularizer, bias_regularizer, activation functions, global average pooling, call back methods with validation on monitoring the validation loss, have been considered to simplify the CNN models and prevent the models from overfitting.

An initial base model with consecutive convolution layers has been constructed. The primary goal was to comprehend how our base CNN model operates. Below is a diagram of the base model architecture:

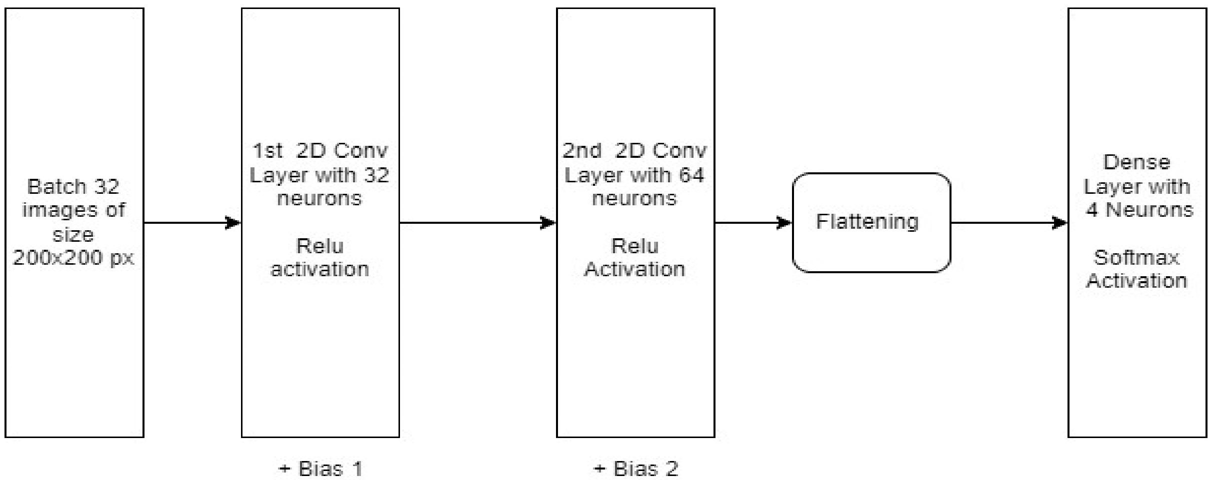

Figure 10: Base Model Architecture

The figure 11 demonstrates base model architecture in which image data passes through 2 convolution layers performing feature extraction and evaluating feature map for next layers. Flattening is used to convert all the resultant 2-Dimensional arrays from pooled feature maps into a single long continuous linear vector. The flattened matrix is fed as input to the fully connected layer to classify the image.

The base model was trained over 50 iterations using callback techniques such as early halting after tracking validation loss for 10 iterations in a row. After the model was created, its performance was assessed using a generator on test data, accuracy, and loss on validation, and a plot of the training and validation data. Basic data augmentation techniques have been used for the underlying model like rotation and flipping.

On the sequential model, regularization approaches have been used to prevent overfitting and increase model generalizability. Following the convolution layer, batch normalization is used to quickly process and normalize the output of the preceding layer, accelerating and stabilizing deep learning. Batch normalization lessens internal covariate shift, which makes training less difficult [40]. In batch normalization, mean and variance are determined for each batch during training, whereas the network uses the moving mean and variance of all images during inference [40]. Dropout has also been employed to speed up the model's learning. During training, it turns off the number of neurons. Both the incoming and outgoing signals are turned off when the neurons are turned off. Allowing each neuron to learn from the training data is the primary goal of the dropout regularization procedure [41].

This approach is utilized in the project with dropout levels of 0.2, respectively, meaning that 20% of the neurons are randomly turned off while the convolution process is being run. Since the trained network's weights have already been saved, dropout regularization is not employed while testing on test data [41]. In order to introduce non-linearity to each neuron, activation functions like SoftMax have been used in this project. By changing weights in backward propagation, activation functions assist in decreasing the discrepancy between real output and neuron output during deep learning training [42].

In this study, XGBoost was used to classify three different types of brain tumor following CNN. Utilizing optimizers, the loss function has been reduced Adam (Adaptive Moment Estimation. These regularization strategies have been applied as the best DL techniques that help generalize the model.

By using pre-trained model DenseNet121 to train a CNN + XGBoost model, transfer learning has also been used. Due to a shortage of training data, this strategy is being used. Transfer learning, which uses a model that has already been trained, can produce a decent CNN model while cutting down on training time. Additionally, the transfer learning strategy is substantially more effective for generalization [43]. One of the innovations in deep learning is transfer learning, which works best with tiny datasets.

XGBoost For classification

As a fresh deep learning model for categorization issues, author introduce the Convolutional eXtreme Gradient Boosting technique. ConvXGB combines the strengths of a CNN.[44] With XGBoost, resulting in a state-of-the-art performance and high accuracy, as we will demonstrate. ConvXGB's methodical method for selecting these two models is one of its primary features.

1. XGBoost is a scalable machine learning algorithm for tree boosting that prevents overfitting and is frequently used by data scientists. It excels on its own and has been shown effective in numerous machine learning competitions.
2. Using CNN, a family of automatic feature learning algorithms, we improve clarity.

In addition to the original model, a total of five CNN models have also been trained using practice photos of brain tumor. Different combinations of regularization techniques and data augmentation strategies have been used for these CNN models. Table 3 provides information on these models and their characteristics.

Model Name	Libraries	Model Properties
Base Model	Tensor Flow	Three Convolution Layers, Softmax Activation Function at Dense Layer, Adam Optimizer
CNN_Model2	Keras	Four Convolution Layers with batch normalization and pooling, Softmax Activation Function at Dense Layer, Dropuout of 0.3, Adam Optimizer
CNN_Model3	Keras	Five Convolution Layers with batch normalization and pooling, Softmax Activation Function at Dense Layer, Dropuout of 0.3, Adam Optimizer
CNN3_Model4	Keras	Six Convolution Layers with batch normalization and pooling, Softmax Activation Function at Dense Layer, Dropuout of 0.3, SGD Optimizer
XGCNN	Keras, XGBoost	Densenet121 with batch normalization and pooling, Softmax Activation Function at Dense Layer, Dropuout of 0.2, ADAM Optimizer

Table 3: List of CNN Models

All the model's testing accuracy have been calculated on testing data and it was found that CNN with XGBoost yield best outcome. In the next section more detailed comparison with results is shown.

4. Result and discussion

The thesis involves three subsets of the dataset—training, validation, and experimentation with percentages of 90%, 5%, and 5%, respectively. The findings of the proposed image datasets employing the planned CNN and XGBoost architecture are reported in the following sections. Pre trained model denseNet121 have been used following the dropout, pooling and dense layers with model running twenty five epochs.

With the addition of fundamental data augmentation, five different CNN models have been trained with varying demands on the layers, optimizers, activation functions. On testing datasets, all models have been assessed, and their accuracy and loss have been evaluated by fitting the models mentioned in Table 4.

The loss and accuracy graphs for training the basic model using the standards provided in the table above are displayed in the figures below. For both training and validation, loss continues to decline, but the model trained with 50 epoch's displays accuracy being improved for both.

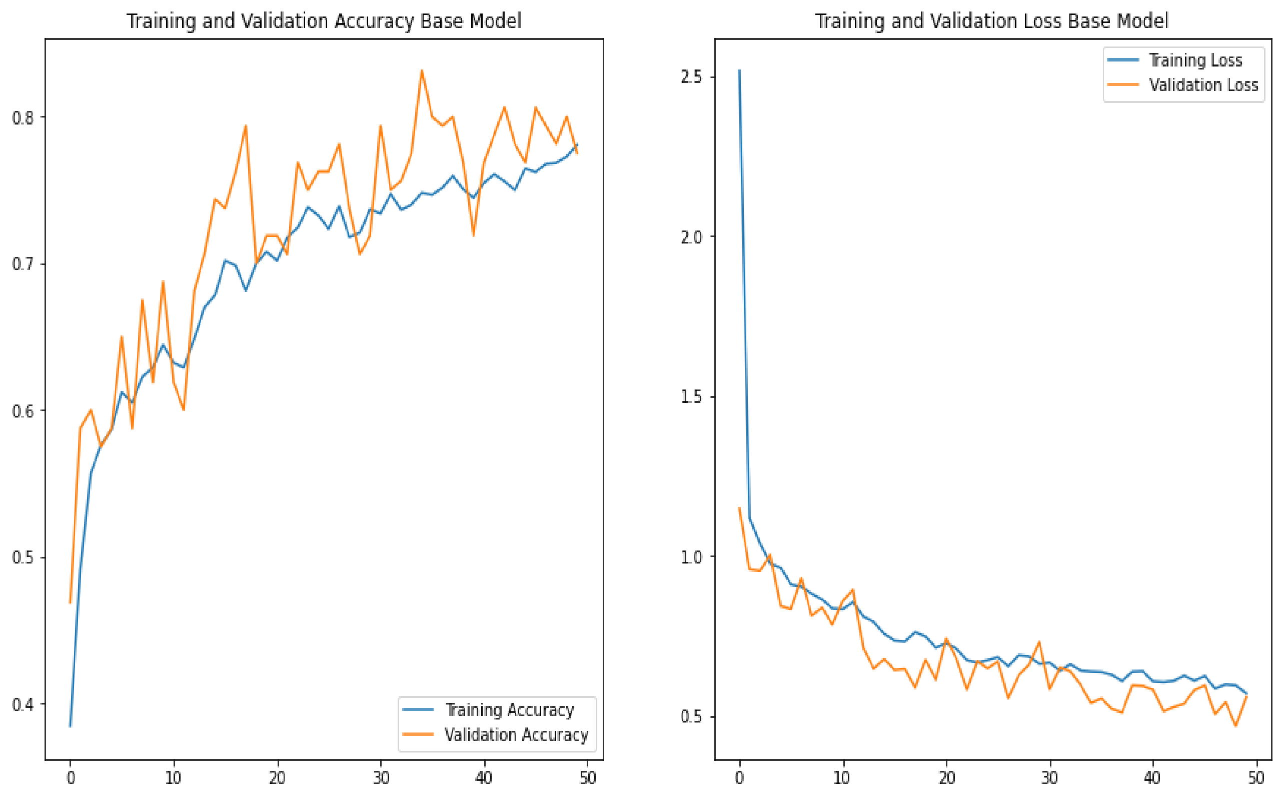

Figure 11: Base Model trained with 50 Epochs

Following the base mode, four other models have been tested, the models (CNN Model2, CNN Model3, CNN Model4, XGBoost) using the parameters listed in Table 3. The other models perform better than the base model resulting in increase in training accuracy, while validation accuracy and loss show significant volatility. This is because each model have more layer than the other irrespective of same augmentation techniques and number of epochs (XGBoost Exception).

Below table shows model specification:

Model	Model Properties
Base Model	Three Convolution Layers, Softmax Activation, Dense Layer, Adam Optimizer
CNN_Model_2	Four Convolution Layers, pooling, Softmax Activation, Dense Layer, Dropout layer of 0.2, Adam Optimizer
CNN_Model_3	Five Convolution Layers, pooling, Softmax Activation, Dense Layer, Dropout layer of 0.2, Adam Optimizer
CNN_Model_4	Six Convolution Layers with batch, pooling, Softmax Activation Function at Dense Layer, Dropout layer of 0.2, ADAM Optimizer
XGCNN	Densenet121 with batch normalization and pooling, Softmax Softmax Activation, Dense Layer, Dropout layer of 0.2, ADAM Optimizer

Table 4: CNN Models

Table 4 demonstrates that the pre-trained model DenseNet121 outperformed other trained models, providing the best results on test data with an accuracy of 98.02%.

Table 5 shows the accuracy score for each tumor category for all the trained models:

Model Name	Pituitary Tumor	Meningioma Tumor	Glioma Tumor
Base Model	66%	52%	71%
CNN_Model_2	79%	46%	61%
CNN_Model_3	69%	41%	80%
CNN_Model_4	71%	62%	88%
XGCNN	99%	98%	97%

Table 5: Accuracy Score on Each Tumor Category

This accuracy rating reflects how effectively the model was able to recognize the type of tumor, making it more of a recall score. Subtract the number of images from the total number of images in that category that were studied. According to the above table, meningioma brain tumor photos are harder to come by than those of other tumor types.

The line charts for testing and validation accuracy and validation loss for our chosen model, XGBoost, are shown below.

Final Model : CNN + XGBoost

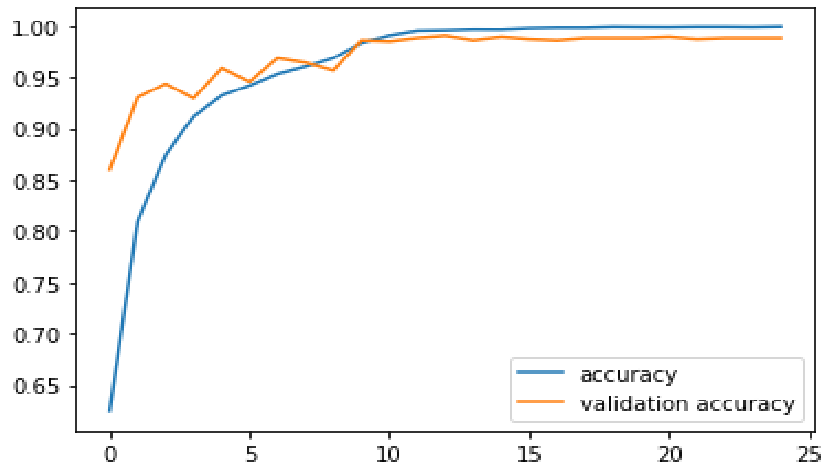

Figure 12: Testing and Validation accuracy plot

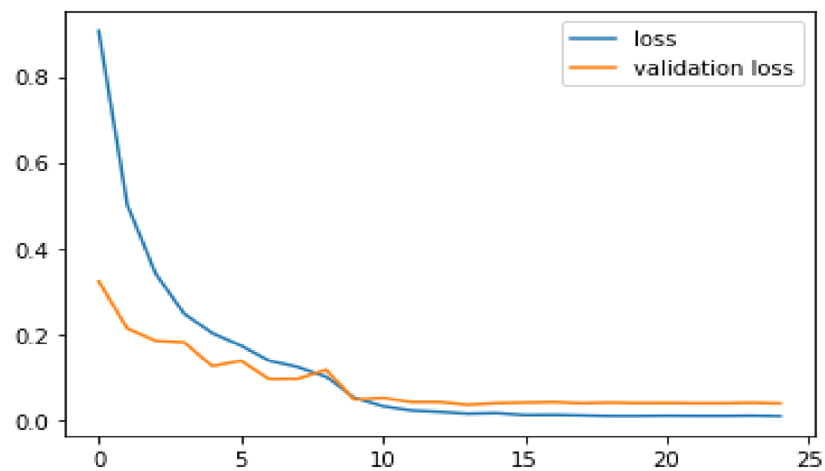

Figure 13: Testing and Validation loss plot

4.1. Main Findings

The primary goal of this research study was to use CNN and several libraries, frameworks and tools to classify photos of 3 different classes of brain tumor. The goal and purpose of this project have been met, and it has been discovered that CNN when used with XGBoost outperforms other models to develop generalize models. This is done by using the pre-trained model DenseNet121 along with a max-pooling and dense layers.

Since it takes a significant amount of time and effort to manually classify an MRI of a brain tumor, radiologists may use this as a complimentary tool to classify brain tumor images more rapidly and effectively by utilizing this classification model. This could be employed as an additional technique for CAD tool as well to aid in the early identification and classification of brain tumor. As early diagnosis can save thousands or more lives, this can significantly alter the outcome.

4.2. Recommendations

The dataset taken into consideration for use have been described throughout the thesis. Finding an equal amount of information or images of various classes of brain tumor from the medical community is very difficult in the real world. Class data balancing techniques, such as SMOTE, oversampling, undersampling, can be utilized prior to pre-processing the picture data and evaluating the model. Additionally, model evaluation metrics like accuracy, recall, F1-score, and AUC curve should be taken into consideration for the generalization of the model if the data used to evaluate the model in practice is uneven. In the case of an uneven dataset like in our case we have different number of images for the respective classes of tumor, the F1 score should be optimum.

4.3. Further Scope

There is always a scope of further advancement in any field of research. Below are the few points that author find could be implemented. The outcomes can be enhanced further by Increase the dataset's size for the model to see more enriched data samples, train it for more epochs, cross-validation, the model can be tuned using Keras Tuner Hyper parameters.

When the data provided is huge, a convolution neural network performs better and produces a better generalized model. There are various available augmentation techniques like mixup, cutmix can be used to train model more effectively. Another technique for building a data set from a little amount of data is N-Shot Learning, also recently, a technique known as "auto augmentation" was also developed to identify the best data augmentation process, enabling the generation of unlimited amounts of data from modest amounts of data. Previous researchers have used some of the techniques mentioned above but not all to improve their work.

This thesis can be taken ahead by utilising other existing models and some changes to the models like additional layers.

5. Conclusion

Three different forms of brain cancers, including gliomas, meningiomas, and pituitary tumor, are included in the dataset used for this study. In this study, the proposed CNN with XGBoost is used to carry out an effective automatic categorization of brain tumor. The information has been processed in a variety of ways, including segmented, cropped, and uncropped tumor. In conclusion, we were able to demonstrate the significance of early brain tumor detection. As a result, this aids in avoiding radiation planning and early brain tumor diagnosis. We were able to do this by providing a thorough introduction to the study, followed by full documentation of the literature review and detailed explanations of the research's objectives, methodology, and risks. In this study effort, a thorough examination of the operation of deep neural networks and their optimization techniques has been conducted.

However, the main challenge in this field of study is locating the appropriate data. It is extremely difficult to find various types of brain tumor photographs in the medical field.

In order to classify and construct a generalize CNN model, XGBoost is utilised. In the medical field, improved brain imaging might help in using CNN to more precisely diagnose different types of brain cancers. It was observed that basic data augmentation using the transfer learning methodology and the DenseNet121 model outperformed other CNN models and techniques previously described.

In this study, we provide a novel CNN in conjunction with XGBoost architecture for grading a brain tumor. Our design was successful in efficiently classifying the three types of brain tumor. Using

T1 weight contrast-enhanced brain MR images, the system can effectively grade the tumor into three levels: meningioma, glioma, and pituitary tumor.

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